Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

REC_2028_OOPS using Java_Week 4_CY

Attempt : 1
Total Mark : 40
Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

A library wants to analyze book titles to count the number of words that start with an uppercase letter. This helps the library track proper nouns and important words in titles.

Your task is to write a program that, for each given title, counts and prints the number of words that start with an uppercase letter.

Input Format

The first line contains an integer T, representing the number of book titles.

Each of the next T lines contains a single title (string).

Output Format

For each title, the output print a single integer representing the number of words starting with an uppercase letter.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1
The Chronicles of Narnia

Output: 3

Answer

```
// You are using Java
import java.util.Scanner;

class CountUppercaseWords {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int T = scanner.nextInt();
        scanner.nextLine();
        for (int i = 0; i < T; i++) {
            String title = scanner.nextLine();
            String[] words = title.split(" ");
            int count = 0;
            for (String word : words) {
                if (word.length() > 0 && Character.isUpperCase(word.charAt(0))) {
                    count++;
                }
            }
            System.out.println(count);
        }
        scanner.close();
    }
}
```

Status : Correct ***Marks : 10/10***

2. Problem Statement

Meera is practicing her English vocabulary. She wants to focus on words that have more vowels in them, as they help improve her pronunciation. She decides to extract only those words from a sentence that contain at least two vowels.

Your task is to help Meera by writing a program that finds such words from the given sentence.

Input Format

The input contains a string representing the sentence.

Output Format

The output prints all the words that contain at least two vowels, separated by a space.

If no such word exists, print "No words with two vowels".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: This is an example sentence
Output: example sentence

Answer

```
// You are using Java
import java.util.Scanner;

class VowelWordsExtractor {
 public static void main(String[] args) {
  Scanner scanner = new Scanner(System.in);
  String sentence = scanner.nextLine();
  String[] words = sentence.split(" ");
  StringBuilder result = new StringBuilder();
  for (String word : words) {
   int vowelCount = 0;
   for (char c : word.toLowerCase().toCharArray()) {
    if ("aeiou".indexOf(c) >= 0) {
```

```
      vowelCount++;
     }
    }
    if (vowelCount >= 2) {
     result.append(word).append(" ");
    }
   }
   if (result.length() > 0) {
    System.out.println(result.toString().trim());
   } else {
    System.out.println("No words with two vowels");
   }
   scanner.close();
  }
}
```

Status : Correct ***Marks : 10/10***

3. Problem Statement

Anjali is preparing a report on text complexity. She wants to identify all words in a sentence that contain at least one digit so she can analyze numeric mentions.

Your task is to write a program that extracts and prints all words containing at least one digit from a given sentence.

If no such word exists, print "No words with digits found".

Input Format

The input contains a single line containing a sentence with multiple words.

Output Format

The output prints all words containing at least one digit separated by a space.

If no word contains a digit, print "No words with digits found".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: The model X100 and Y200 are available
Output: X100 Y200

Answer

```
import java.util.*;

public class Main {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      String sentence = sc.nextLine().trim();
      String[] words = sentence.split(" ");
      StringBuilder result = new StringBuilder();

      for (String word : words) {
         if (containsDigit(word)) {
            result.append(word).append(" ");
         }
      }

      if (result.length() > 0) {
         System.out.println(result.toString());
      } else {
         System.out.println("No words with digits found");
      }
      sc.close();
   }

   private static boolean containsDigit(String word) {
      for (char c : word.toCharArray()) {
         if (Character.isDigit(c)) return true;
      }
      return false;
   }
}
```

Status : Correct ***Marks : 10/10***

4. Problem Statement

A bookstore wants to analyze the titles of books to determine their longest word in each title. This helps in designing banners and covers.

Your task is to write a program that, given a sentence (book title), finds and prints the longest word. If multiple words have the same maximum length, print the first one.

Input Format

The input contais a single line containing a sentence representing the book title.

Output Format

The output prints a string representing the longest word in the sentence (book title).

Refer to the sample output for formatting specifications.

Sample Test Case

Input: The Chronicles of Narnia
Output: Chronicles

Answer

```
// You are using Java
import java.util.Scanner;

 class LongestWordFinder {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String title = scanner.nextLine();
        String[] words = title.split(" ");
        String longestWord = "";

        for (String word : words) {
            if (word.length() > longestWord.length()) {
                longestWord = word;
            }
        }
```

```
        System.out.println(longestWord);
        scanner.close();
    }
}
```

Status : Correct *Marks : 10/10*